

AI infrastructure and distributed systems

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I build and operate the systems behind enterprise AI and data platforms—from telemetry ingestion and cloud migrations to vector retrieval and GPU-backed model serving.

01 Enterprise GenAI platform

Apple · Contract via Mphasis · Apr 2024–Aug 2026

02 Cyber telemetry & distributed ingestion

Cloudwick · Cyber Data Lake / Amorphix · Nov 2016–Dec 2018

03 Mission-critical cloud migration

Cloudwick · Client: Freddie Mac · Jan 2019–Aug 2023

04 Cloud automation, identity & recovery

Cloudwick · Client: Freddie Mac · Jan 2019–Aug 2023

Sanitized reconstructions and career-reported outcomes. Full case studies and supporting diagrams are available on the website.

Enterprise GenAI platform

Apple | Contract via Mphasis | Apr 2024–Aug 2026

I developed and co-owned model onboarding and deployment tooling, productionized ingestion and inference services, and authored the Solr infrastructure-as-code and lifecycle tooling end to end. My responsibility crossed Kubernetes, GPU serving, search infrastructure and production operations.

7.0s → 1.9s

Approximate vector-query latency

\$266K+

Annualized Solr migration savings

250+ nodes

Across vector-database infrastructure

What I owned

- Developed and co-owned Helm/Argo CD model onboarding on a dedicated GPU EKS cluster, integrated with an ingestion and inference platform on a separate EKS cluster.
- Authored and owned Solr IaC and lifecycle tooling: infrastructure provisioning, SolrCloud/ZooKeeper configuration, exporters, replica operations, restart workflows and migration validation.
- Productionized 10+ GenAI microservices across development, UAT and production, including deployment manifests, environment overlays, routing, logging, metrics and release diagnosis.
- Served as an escalation SME for pod failures, memory pressure, storage problems and query latency; connected recurring failures to automation and operational improvements.

Scope: Sanitized logical composite of production configuration families and my account of the platform. It is not a live cluster inventory. Results are reported career outcomes; no LLM throughput, TTFT or GPU-utilization improvement is claimed. Solr authorship does not imply sole authorship of the broader platform or upstream components.

Engineering decisions

Separate application and model infrastructure

The architecture separates the ingestion/inference services from GPU model hosting. This lets application and model deployments have distinct capacity and runtime configurations.

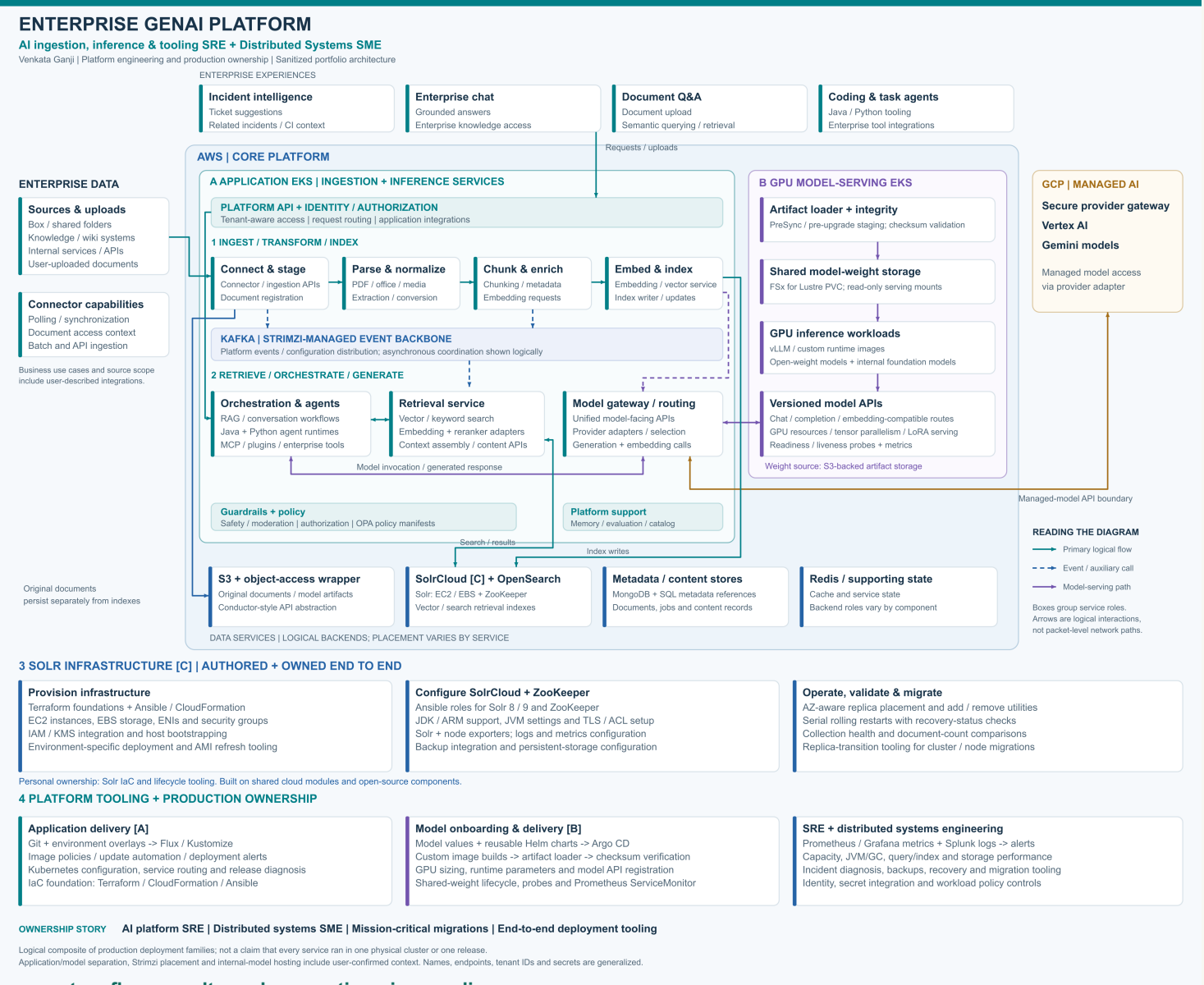
Make model artifacts part of deployment

Weight staging, integrity checks, shared volumes, health probes and runtime values are first-class deployment steps. Configuration covers precision, context limits, memory budgets, tensor parallelism and LoRA serving.

Treat retrieval as a systems problem

I worked across JVM/GC, OS settings, cache behavior, query patterns and indexes. The reported latency improvement is a vector-query result, not an LLM generation benchmark.

Enterprise GenAI platform



Explore the full-size diagram, system flow, results and supporting views online →

Cyber telemetry & distributed ingestion

Cloudwick | Cyber Data Lake / Amorphic | Nov 2016–Dec 2018

I helped build the cyber data platform, implemented multi-source telemetry collection, designed a three-rack capture/replay test bed, and owned deployment and performance work across the hardware and distributed-data stack.

3 racks

Capture and integration test bed

40 Gb/s

Packet-capture technology tuned

4 source types

PCAP, flow, app and firewall telemetry

What I owned

- Implemented collection of PCAP, NetFlow, application logs and firewall events to support a correlated cyber data lake.
- Designed a three-rack test bed with packet brokers and IDS/IPS integrations; deployed and performance-tuned 40 Gb/s packet-capture technology.
- Engineered Corelight and IXIA integrations and operated Hadoop, Kafka, Spark, Solr, HBase and Elasticsearch components supporting the product.
- Built Prometheus monitoring for collector agents and automated cloud/on-premises deployment using CloudFormation, Ansible, Jenkins, GitLab, Kickstart, Puppet and Python APIs.

Engineering decisions

Separate functional tests from performance tests

A functionally correct collector can still be limited by disk, broker capacity or the capture path. The test environment made those dimensions visible instead of treating every failure as an application problem.

Keep capture and management paths distinct

The multi-rack design separates the data-transfer network from server-management access. Capture/replay and security integrations can be reasoned about independently from routine host operations.

Observe the collector as well as the cluster

Prometheus monitoring at the collector layer complements distributed-platform operations. This gives the team visibility before data reaches the search or analytics tier.

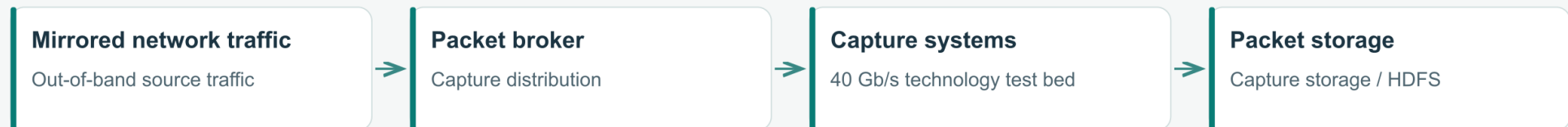
Scope: The public diagrams are simplified redraws. Gateway capacities and test configurations are not promoted to production benchmarks. The separate Amorphic cloud reference is dated June 2019 and credits a team author; it is included only as product context, not as evidence that I personally designed that entire later architecture. Telemetry here does not mean OpenTelemetry.

Cyber telemetry & distributed ingestion

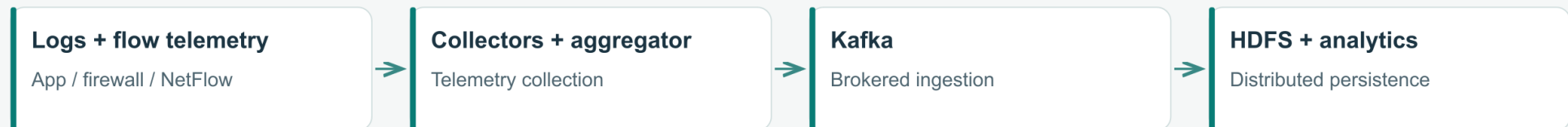
Cyber telemetry and distributed ingestion

Logical platform roles across capture, collection, storage and investigation

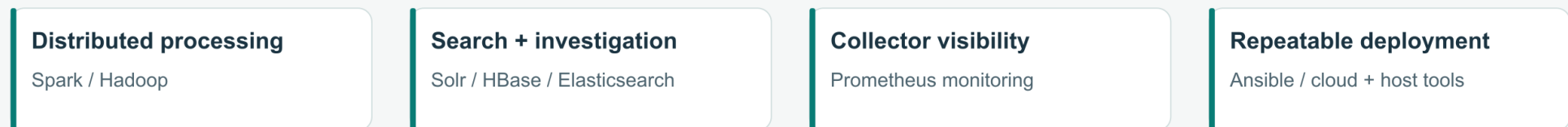
PACKET PATH



EVENT PATH



CROSS-CUTTING CAPABILITIES



Rows group platform capabilities; they do not assert one pipeline for every data source.

Explore the full-size diagram, system flow, results and supporting views online →

Mission-critical cloud migration

Cloudwick | Client: Freddie Mac | Jan 2019–Aug 2023

I led mission-critical cloud migration work and designed the transition of a mortgage-document platform from Cloudera Hadoop and Solr to AWS object storage and distributed search. I led Spark migration development and coordinated infrastructure, data movement, testing and production transition.

3 migrations

Mission-critical programs led in the engagement

3–8 engineers

Approximate teams technically led

HDFS → S3

Document-storage modernization

What I owned

- Led three mission-critical migrations across distributed-data and live application systems during the engagement, coordinating architecture, automation, data movement, performance testing and production transition.
- Designed and productionized a document-platform migration from Cloudera Hadoop/Solr to AWS Elasticsearch/OpenSearch, ELK and S3.
- Led development of Spark migration jobs moving Solr data into Elasticsearch/ELK and HBase data into S3.
- Built reusable infrastructure and operations automation and partnered with test teams to performance-test infrastructure at scale using Dynatrace.

Engineering decisions

Separate transport from transformation

DistCp can address bulk transfer, while archive extraction and target naming/mapping require additional work. The design makes those responsibilities explicit.

Keep document and search migrations coordinated

S3 objects and search records serve different roles. The migration program covered both, so the target architecture could support application behavior as well as storage.

Include operations in the migration scope

Provisioning, performance testing, snapshots and recovery were part of the engineering work. A cloud cutover needed an operating model as well as a target diagram.

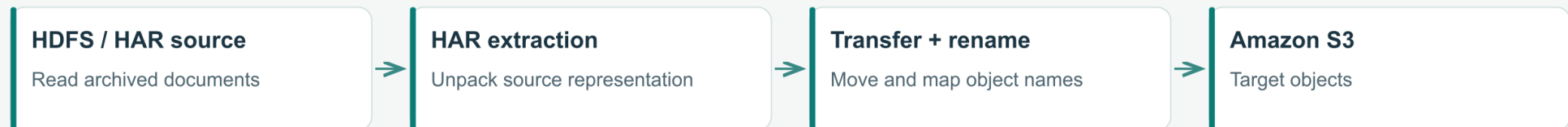
Scope: The HAR diagram presents alternatives. It does not by itself establish the selected route for every workload, total bytes migrated, zero downtime or a benchmark. The leadership and delivery claims come from my career record; the drawing explains the technical approach.

Mission-critical cloud migration

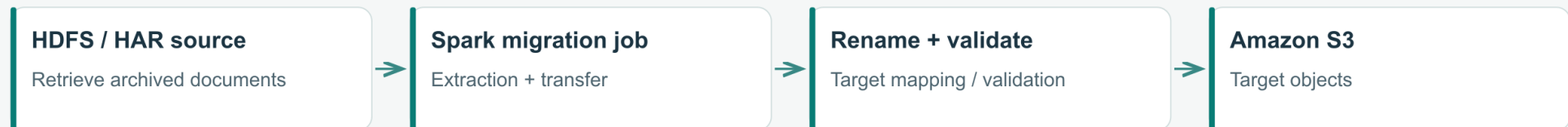
Hadoop archives to cloud object storage

Migration approaches shown in the design; separate from the search-data migration

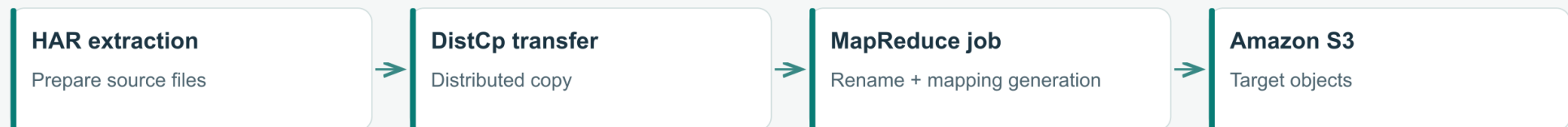
STAGED APPROACH



CUSTOM SPARK APPROACH



DISTCP + MAPREDUCE APPROACH



The source drawing compares approaches; it does not establish one chosen route for every workload.

Explore the full-size diagram, system flow, results and supporting views online →

Cloud automation, identity & recovery

Cloudwick | Client: Freddie Mac | Jan 2019–Aug 2023

I designed and productionized cloud infrastructure for a document and search platform, including proxy access, federated authentication, reusable IaC and hybrid-cloud disaster recovery. The work joined application requirements with security and operational constraints.

~15 min

RPO for the delivered DR environment

~2 hours

RTO for the delivered DR environment

4 environments

Federated authentication scope

What I owned

- Designed proxy, load-balancing and autoscaling infrastructure supporting the document platform's move to AWS.
- Developed reusable Terraform, CloudFormation and Ansible provisioning/landing-zone workflows covering compute, search, networking, IAM, secrets, encryption and cross-region storage.
- Designed federated authentication using PingFederate, Vault, AWS IAM/STS, ADFS and SAML across four environments; partnered with InfoSec on on-premises-to-S3 connectivity.
- Built and productionized a hybrid-cloud DR environment using cross-region replication, S3, Airflow, Python and Elasticsearch/ELK, delivered within six months.

Scope: Drawings represent different releases and environments; the portfolio shows their relationship without implying that every identity mechanism runs in the same request path. Recovery metrics are career-reported results for the described DR environment, not a universal service guarantee.

Engineering decisions

Bridge the legacy application boundary

The proxy tier provides an integration point between the existing application and cloud services. The design also spreads proxies across availability zones and uses autoscaling.

Treat identity as architecture

Directory integration, federation and temporary AWS credentials influence how applications reach the data plane. They were designed alongside network paths, not added as an afterthought.

Make recovery and IaC reinforce each other

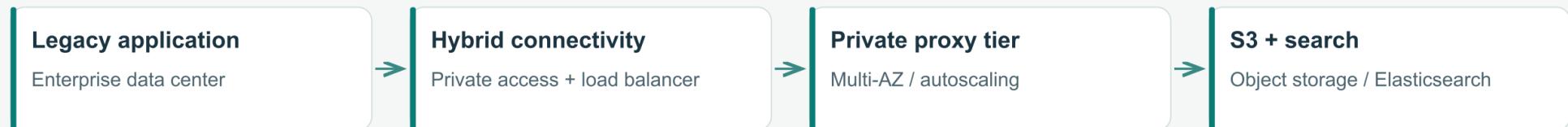
Reusable provisioning defines the target resources; backup/replication provides recoverable state. The architecture distinguishes those responsibilities, including an on-demand recovery footprint.

Cloud automation, identity & recovery

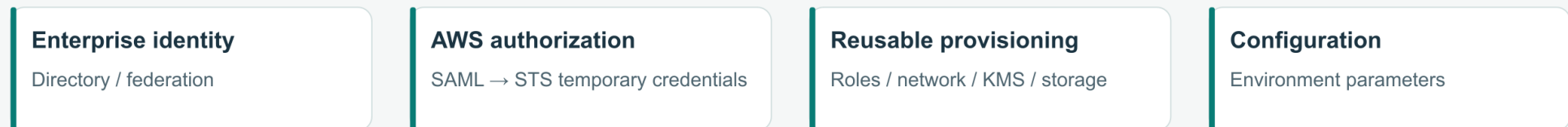
Hybrid cloud document and search platform

Application access, cloud foundations and recovery shown as distinct responsibilities

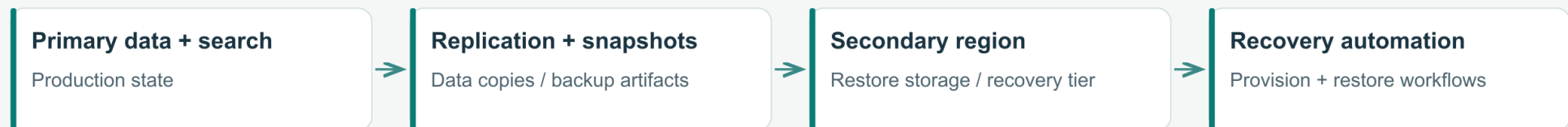
APPLICATION DATA PATH



IDENTITY AND INFRASTRUCTURE



RECOVERY RESPONSIBILITIES



Logical composite of releases. DR results: approximately 15-minute RPO and 2-hour RTO for the delivered environment.

Explore the full-size diagram, system flow, results and supporting views online →